Orthodontics & Dentofacial Orthopedics

Departmental Objectives

At the end of the course, the student should be able to:

1. Identify and diagnose anomalies of the dentition, occlusion, facial structures and abnormal functional conditions (Orthodontic patients)
2. Detect deviations of the development of the dentition, of facial growth, and occurrence of functional abnormalities
3. Identify pernicious oral habits that may lead to mal occlusion.
4. Conduct interceptive orthodontic measures
5. Evaluate need for orthodontic treatment
6. Formulate a treatment plan for simple type of malocclusions and execute
7. Design, plan & fabricate simple universal type of Orthodontic treatment procedures(Hawley type appliances) insert and activate.
8. Design, plan &fabricate functional appliance & insert, learn how to activate.
9. Describe basic concept of fixed orthodontic appliance.
10. Advice & aware the patients to take specialized Orthodontic consultation & refer when necessary.

List of Competencies:

1. Demonstrate basic knowledge and skill to examine, investigate and diagnose the patient's malocclusion for orthodontic treatment.
2. Manage simple Orthodontic problems of patients at primary health care facilities.
3. Identify complicated Orthodontic problems, able to take initial care to them and able to refer to appropriate site for further management without causing any detoriation of patient's condition.
4. Communicate with the patients regarding preventive, curative & rehabilitative orthodontic care.

Distribution of teaching - learning hours

Lecture	Demonstration	Practical	Clinical Teaching	Total Teaching hours	Integrated Teaching (Common)	Formative Exam		Summative exam	
						Preparatory leave	Exam time	Preparatory leave	Exam time
180 hrs	30 hrs	50 hrs	80 hrs	340 hrs	10 hrs	10 days	20 days	10 days	35 days

Marks distribution:

Final professional examination:

Marks distribution

- Total Marks-300

  Pass Marks 60% in each part

  1. Written -100 (SAQ-70, MCQ-20, FORMATIVE-10)
  2. OSPE/OSCE -100 (Practical-50 + Clinical-50)
  3. SOE-100

Orthodontics & Dentofacial Orthopedics

Learning Objectives	Contents	Teaching Hours
• Define Orthodontics & Dentofacial Orthopedics. • Describe background of Orthodontics • Define normal & abnormal occlusion & related definitions • Define ideal occlusion • Define & mention Andrews 6 keys to normal occlusion. • Mention aims, objectives and scope of Orthodontics	1. Introduction • Define Orthodontics & Dentofacial Orthopedics. • Background of Orthodontics • Define normal & abnormal occlusion & related definitions • Ideal occlusion-introduction, definition & Andrews 6 keys to normal occlusion. Aims, objectives and scope of Orthodontics: Facial Aesthetics, Normal function, Stability.	Lecture: 10
• Define primary, mixed & permanent dentition. • Describe dimensional changes in the dental arches during different dentition period. • Describe the normal growth of jaw, teeth & face. • Describe changes in face form & profile. • Mention psychological & Social impact of abnormal growth & malocclusion • To identify normal & abnormal growth pattern of jaws & dentition & their psychological & social impact.	2. Growth & Development of dentitionjaws, palate & face. • Define primary, mixed & permanent dentition. • Dimensional changes in the dental arches during different dentition period. • Describe the normal growth of jaw, teeth & face. • Describe changes in face form & profile. Psychological & Social impact of abnormal growth & malocclusion	Lecture: 12 Demo: 1 clinical: 8
• Describe epidemiology of malocclusion including incidence & prevalence • To find out prevalence of malocclusion & Biostatics.	3. Epidemiology. • Describe epidemiology of malocclusion including incidence & prevalence Biostatics	Lecture: 5
• Describe-Lip morphology (competent, incompetent, everted, hyperactive) & its influence on occlusion. • Explain anatomy & behavior of tongue. • Explain swallowing behaviors • Describe the effects of Adenoids • Explain Breathing and Speech mechanism. • Explain Respiratory sleep apnea. • To evaluate the soft tissues and swallowing pattern and its impact on malocclusion.	4. Soft tissue Morphology and behaviors • Describe-Lip morphology (competent, incompetent, everted, hyperactive)& its influence on occlusion. • Explain anatomy & behavior of tongue. • Swallowing behaviors • Describe the effects of Adenoids • Breathing and Speech. • Respiratory sleep apnea.	Lecture: 10 Demo:1 Clinical: 8

Learning Objectives	Contents	Teaching Hours
• To illustrate normal occlusion, malocclusion with classification & etiology • Classify Malocclusion (Angles & others) • Describe untoward effect Malocclusion if not treated. • Describe different type of Malposition. • State the etiology of Malocclusion -(General & local factors)	5. Malocclusion: Classifications & etiology: • Classify Malocclusion (Angles& others) • Describe untoward effect • Describe different type of Malposition. State the Etiology-(General & local factors)	Lecture: 8 Demo:1 Clinical:4 Practical:1
• Take comprehensive history • Classify malocclusion • Examine teeth& periodontal structure. • Appraise of soft tissue • Perform functional analysis • Plan the necessary investigation • Maintain appropriate diagnostic record • Analyze & interpret the records. • Outline the management protocol • Communicate with the patient to aware the probable prognosis & financial involvement • To perform intra & extra oral examination, interpretation, & prognosis.	6. Diagnosis of Malocclusion • Obtain comprehensive history • Extraoral & intra-oral examination • Identification of malocclusion • Examination of teeth • Appraisal of soft tissue • Functional analysis • Plan the necessary investigation • Maintain appropriate diagnostic record • Analyze& interpret the records. • Outline the management protocol Communicate with the patient to aware the probable prognosis & financial involvement	Lecture: 12 Demo:1 Clinical:8 Practical:4
• To narrate how to evaluate study model to asses tooth-jaw discrepancy, photograph, different radiograph for orthodontic treatment & its prognosis. • Obtain impression & plaster model. • Describe technical procedure for impression & plaster model • Analyses of the study model to assess tooth-jaw discrepancy: arch perimeter, arch length, arch width etc. • Define & interpret OPG • Define cephalogram•	7. Diagnostic Techniques a. Impression technique & preparation of study model • Obtain impression & plaster model. • Technical procedure for impression & plaster model • Analysis of the study model to assess tooth-jaw discrepancy: arch perimeter, arch length, arch width etc. b. Intraoral & Facial photograph. c. Intra oral radiograph d. Extra oral Radiograph	Lecture: 12 Demo:4 Clinical: 8 Practical:5

Learning Objectives	Contents	Teaching Hours
● Define cephalometry ● Mention anthropological sources &describe development of cephalometrics ● Mention objectives of cephalometric tracings ● Identify cephalometric landmarks –Cranial, Maxillary & Mandibular ● Perform cephalometric Analysis-Dental, Skeletal, & Skeletal-Dental analysis	● OPG ● Cephalogram ➘ Define cephalometry ➘ Anthropological sources & development of cephalometrics ➘ Objectives of cephalometric tracings ➘ Cephalometric landmarks –Cranial, Maxillary & Mandibular ➘ Cephalometric Analysis-Dental, Skeletal & Skeletal-Dental analysis ➘ Tracing of self cephalogram to compare with Bangladeshi norms (Stainer analysis) a. CBCT(Cone-beam Computed tomography)	
● Summarize a general concept about orthodontic tooth movement. ● Describe different Tissue change ● Differentiate physiologic movement from orthodontic movement. ● Describe Patho-physiological change of tissues. ● Describe Histopathological changes at the pressure & tension area. ● Mention types of tooth movement. ● Describe theory of tooth movement. ● Explain effect of normal and excessive force ● Explain the tissue changes with different type of appliance ● Explain Biological basis of Orthodontics Therapy. ● State favorable and unfavorable incidence of tooth movement.	8. Tissue Changes & Tooth movement ● Describe different Tissue change ● Difference between physiologic movement and orthodontic movement. ● Describe Patho-physiological change of tissues. ● Histopathological changes at the pressure & tension area. ● Types of tooth movement. ● Theory of tooth movement. ● Explain effect of normal and excessive force ● Explain the tissue changes with different type of appliance ● Explain Biological basis of Orthodontics Therapy. State favorable and unfavorable incidence of tooth movement.	Lecture: 9 Demo:2 Practical:2

Learning Objectives	Contents	Teaching Hours
• Explain preventive Orthodontics & Methods. • Describe interceptive Orthodontics & Methods. • Explain serial extraction, space maintainer tongue guard & habit breaking appliances. • Narrate Growth regulatory appliances.	9. Preventive & Interceptive Orthodontics • Explain preventive Orthodontics & Methods. • Describe interceptive Orthodontics & Methods. • Explain serial extraction, space maintainer tongue guard & habit breaking appliances. Narrate Growth regulatory appliances.	Lecture: 8 Demo:1 Clinical: 4 Practical:2
• Select the patient according to orthodontic treatment need.	10. Consideration of Orthodontic Treatment needsaccording IOTN & other orthodontic indexes.	Lecture: 5 Clinical: 4 Practical :2
• Describe force & mechanics in relation to orthodontics. • Describe Force, stress, strain, translation, centre of resistance & centre of rotation. • Describe principles of Orthodontic force control.	11. Bio- Mechanics of tooth movement. • Force,stress, strain, translation, center of resistance & centre of rotation. Describe principles of Orthodontic force control.	Lecture: 9 Demo:1 Practical:1
• Mention different materials & instruments used in orthodontics. • Describe Impression & model materials. • List different type of resin. • Mention different orthodontic bonding & cementing materials. • Describe properties of S.S, Ni-Ti wire & recently developed wires. • Define soldering • Describe composition & properties of silver Solder & Fluxes. • Describe soldering method & procedure. • Define welding • Describe Principle & mechanism of spot welding. • Describe heat treatment procedure. • To evaluate about orthodontics materials & instrument.	12. Materials, instruments used in orthodontics. • Specify different materials & instruments used in orthodontics. • Impression & model materials. • Different type of resin. • Different orthodontic bonding& cementing materials. • Properties of S.S, Ni-Ti wire & recently developed wires. • Soldering- Introduction, definition Composition & properties of silver Solder & Fluxes. • Soldering method & procedure. • Welding-Definition, Principle & mechanism of spot welding. • Heat treatment procedure.	Lecture: 8 Demo:3 Practical:7

Learning Objectives	Contents	Teaching Hours
• Describe different types of anchorage. • Describe preparation and assessment of anchorage planning. • Describe anchorage planning according to need-Mild, Moderate & Maximum • Mention uses of headgear, Chin cap & other Extra-oral /Intra-oral anchorage. • Describe temporary anchorage device (TAD) • Summarize the need, type & value of anchorage.	**13. Anchorage** • State types. • Preparation and assessment of anchorage planning. • Anchorage planning according to need-Mild, Moderate & Maximum • Increase anchorage value- Uses of headgear, Chin cap & other Extra-oral /Intra-oral anchorage. Temporary anchorage device (TAD)	Lecture: 9 Demo:1 Clinical: 4
• Diagnose of simple & complex malocclusion. • Describe procedure of planning of extraction & non-extraction • Describe treatment of class I, II & III malocclusion with certain aims & objectives. • Identify malocclusion & their management	**14. Management of different Malocclusion with different appliance system:** • Diagnosis: Diagnosis of simple & complex malocclusion. • Planning: Planning of extraction & non-extraction • Treatment: Treatment of class I, II & III malocclusion with certain aims & objectives.	Lecture: 9 Demo:1 Clinical: 4 Practical:2
• Define removable appliance • Mention basic requirement for a removable orthodontic appliance. • Perform General wire bending exercise • List components of removable appliance • Design & construct different springs & clasps. • Describe general principle of design and fabrication of removable appliance. • Mention types of appliance for different tooth movement, eg. labiolingual, expansion & contraction of arches • Construction of Hawley, Begg's retainer & Bite planes • Perform Trimming & polishing. • Provide Insertion advice & interaction for patients. • Follow up & adjust	**15. Removable appliance- Technique & training.** • Definition. • Basic requirement for a removableorthodontic appliance. • General wire bending exercise • Component of removable appliance • Design &construction of different springs & clasps. • Describe general principle of design and fabrication of removable appliance. • State type of appliance for different tooth movement, eg. labiolingual, expansion & contraction of arches • Construction of Hawley, Begg's retainer & Bite planes • Trimming & polishing. • Insertionadvice& interaction for patients. • Follow up & adjustments	Lecture: 10 Demo:4 Clinical: 4 Practical:9

Learning Objectives	Contents	Teaching Hours
• To evaluate about functional &Orthopedics appliance • Describe Orthopedic force & its Principles • Narrate myo-functional appliance & describe its indication &contraindication • Describe technique & training for construction of Myo-functional appliance • Describe clinical & laboratory steps in construction of Class-II & Class-III activator (Anderson /Mono block type) • Adjust of activator after insertion in the oral cavity	**16. Functional appliance & dento-facial orthopedics** • Describe Orthopedic force & its Principles • Narrate myo-functional appliance & describe its indication & contraindication • Technique & training for construction of Myo-functional appliance • Clinical & laboratory steps in construction of Class-II & Class-III activator (Anderson /Mono block type) • Adjustment of activator after insertion in the oral cavity	Lecture: 11 Demo:3 Clinical: 4 Practical:6
• Describe Principles, identify parts and appliance system currently used. • List the advantages and disadvantages of Fixed Appliance • Describe Technique & training of fixed appliance. • Perform General wire bending exercise • Mention uses of multiple loops used in fixed appliance. • Describe upper & Lower ideal arch formation • Describe Offset & inset bend, 1st,2nd& 3rd order bend ,Toe in & Tip back bend etc. • Describe process of Molar Band formation & welding of molar tube in the band in ideal position. • Able to perform Cementing of the band. • Describe Weldable bracket positioning • Describe direct bonding technique of mesh bracket. • Adjustof arch wire.	**17. Fixed Appliance- Technique & training Elementary knowledge** • Describe Principles, identify parts and appliance system currently used. • List the advantages and disadvantages. • Technique & training of fixed appliance. • General wire bending exercise • Use of multiple loops used in fixed appliance. • Upper & Lower ideal arch formation • Offset & inset bend, 1st,2nd& 3rdorder bend,Toe in & Tip back bend etc. • Molar Band formation& welding of molar tube in the band with ideal position. • Cementing of the band. • Weldable bracket positioning • Direct bonding technique of mesh bracket. • Adjustment of arch wire.	Lecture: 6 Demo:2 Clinical: 4 Practical:5

Learning Objectives	Contents	Teaching Hours
• Differentiate the sequence of fixed appliance	**18. Fixed Appliance- Technique & training Elementary knowledge.** • Stages of treatment progression by fixed appliance: Anchorage planning, Leveling, Canine retraction, Arch / Anterior	Lecture: 8 Demo:1 Clinical: 4
• Identify retention & relapse & overcome procedure • Describe retention & relapse • Evaluate relapse after Orthodontic treatment. • State different type of retainer.	**19. Retention and relapse** • Retention & relapse • Relapse after Orthodontic treatment. • Different type of retainer.	Lecture: 9 Demo:1 Clinical: 4
• Evaluate Multi-disciplinary approach in treating CLP, other endodontic & restorative procedure.	**20. Orthodontic: Multi-disciplinary approach** • Multi-disciplinary treatment procedures. • Cleft Palate management • Pre-surgical Oral-orthopedic & Orthodontic procedure, Post-surgical Orthodontic procedure. • Pre-restorative Orthodontics procedure. • Orthodontics in relation to periodontal disease.	Lecture: 5 Demo:1 Clinical: 4 Practical:2
• Illustrate different appliance technique in adult • Describe adult orthodontics. • State appliance and Technique for adult orthodontics: lingual orthodontics, invisalign orthodontics.	**21. Adult Orthodontics** • Describe adult orthodontics. • State appliance and Technique for adult orthodontics: lingual orthodontics, invisalign orthodontics. • Surgical orthodontics.	Lecture: 5 Demo:1 Clinical: 4 Practical:2

Teaching methods				Teaching aids	In course evaluation
Large group	Small group	Self learning	Others		
-Lecture -video presentation	1. Practical & tutorial:- Demonstration & preparationof different orthodontic appliance in OPD. 2.Clinical:- chair side teaching & performing clinical examination (Problem based learning)	Assignment Self Model study Wire bending Appliance fabrication Appliance design Etc,	Integrated	-Black board & chalk -Whiteboard &Marker -Transparency & marker -computer -CD -Laptop, Multimedia -Flip chart -Slide projector -X-ray plate & viewer -Tracer -Specimen _Analyze report -Model etc.	-Item examinations: SOE -4 Card final examinations: (SOE) -2 Assessment Examinations: Written, SOE, Clinical/practical final (OSPE/OSCE) -Final Examination Written, SOE, Clinical/practical final (OSPE/OSCE)